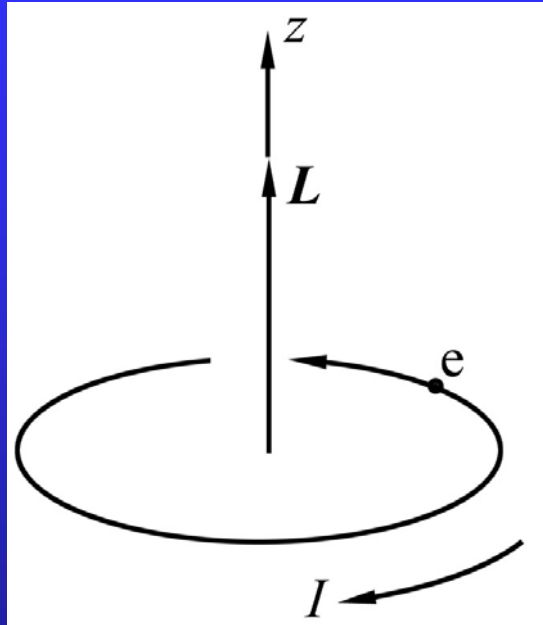


Chapter 19 Magnetic Properties of Matter

19.1 Magnetic dipole of microscopic particles



$$\mathbf{L} = \mathbf{r} \times \mathbf{p} = r m_e v \mathbf{k}$$

$$\boldsymbol{\mu}_L = I \pi r^2 (-\mathbf{k})$$

$$= \frac{ev}{2\pi r} \pi r^2 (-\mathbf{k})$$

$$= \frac{e}{2m_e} m_e v r (-\mathbf{k})$$

$$\mu_L = -\frac{e}{2m_e} \mathbf{L}$$

$$= \gamma \mathbf{L}$$

$$= -\frac{e\hbar}{2m_e} \frac{\mathbf{L}}{\hbar}$$

$$= -\mu_B \frac{\mathbf{L}}{\hbar}$$

The gyromagnetic ratio

$$\gamma = -\frac{e}{2m_e}$$

$$\mu_B = \frac{e\hbar}{2m_e} = 9.27 \times 10^{-24} \frac{\text{J}}{\text{T}}$$
$$= 9.27 \times 10^{-24} \text{ A} \cdot \text{m}^2$$

Bohr magneton

the first Bohr radius

$$\langle r \rangle = a_0 = 4\pi\epsilon_0 \frac{\hbar^2}{me^2} = 0.529 \times 10^{-10} \text{ m} \approx 0.5 \text{ \AA}$$

$$L = a_0 m v = \left(4\pi\epsilon_0 \frac{\hbar^2}{me^2} \right) m \left(\frac{1}{4\pi\epsilon_0} \frac{e^2}{\hbar c} \right) c$$
$$= \hbar$$

$$\mu_L = \mu_B \frac{L}{\hbar} = \mu_B$$

‘old quantum theory’

In quantum mechanics, the angular momentum is quantized:

$$L_z = m\hbar,$$

$$L^2 = l(l+1)\hbar^2.$$

In 1926 Uhlenbeck and Goudsmit: electron spin angular momentum $\mathbf{S}$. --- intrinsic property of the electron

$$S_z = \pm \frac{1}{2} \hbar$$

+ up spin

– down spin

$$\mu_S = -\frac{e}{m_e} \mathbf{S}$$

$$\gamma = -\frac{e}{m_e}$$

$$= -2\mu_B \frac{\mathbf{S}}{\hbar}$$

Note $\gamma = -\frac{e}{2m_e}$ for ‘orbital’ motion.

The nucleons proton
neutron also have magnetic moments.

nuclear Bohr
magneton

$$\mu_N \equiv \frac{e\hbar}{2m_p} = 3.15 \times 10^{-8} \frac{\text{eV}}{\text{T}} \ll \mu_B$$

The spin of the nucleons: $S_z = \pm \frac{1}{2} \hbar$

$$\mu_p = +2.792847386 \mu_N$$

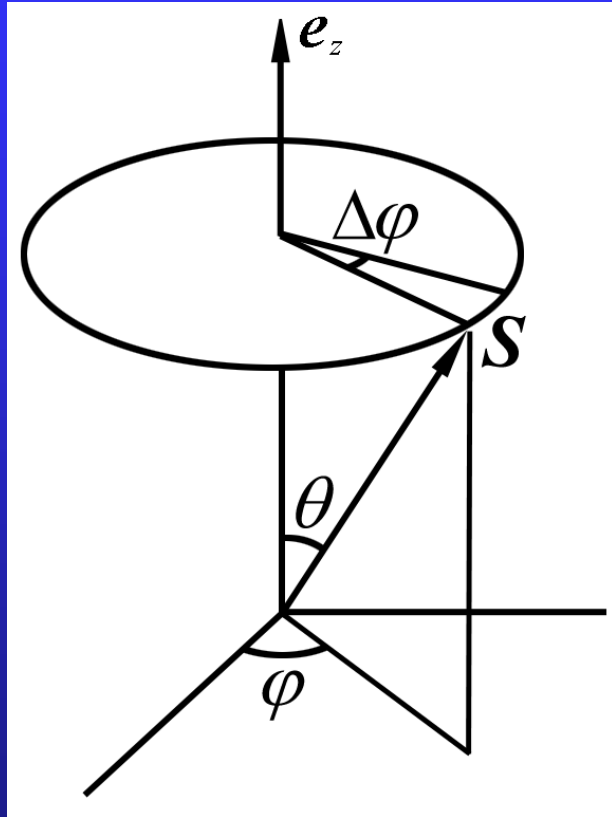
$$\mu_n = -1.91304275 \mu_N$$

the anomalous
magnetic moment

not equal to $2\mu_N \frac{S_z}{\hbar}$

Precession of spin angular momentum

In external magnetic field $\mathbf{B} = B \mathbf{e}_z$,



$$\mathbf{M} = \boldsymbol{\mu} \times \mathbf{B}$$

$$\Omega_p = \frac{\Delta\phi}{\Delta t}$$

$$\Delta\phi = \frac{\Delta S}{S \sin \theta} = \frac{M \Delta t}{S \sin \theta} = \frac{\mu B \sin \theta \Delta t}{S \sin \theta}$$

$$\Omega_p = \frac{\mu B}{S} = |\gamma| B$$

$$\mathbf{M} = \boldsymbol{\mu} \times \mathbf{B}$$

$$\boldsymbol{\mu} = \gamma \mathbf{L}$$

$$\frac{d\mathbf{L}}{dt} = \mathbf{M} = \boldsymbol{\mu} \times \mathbf{B} = \gamma \mathbf{L} \times \mathbf{B} = -\gamma \mathbf{B} \times \mathbf{L}.$$

Similar to the gyroscopic motion, the frequency of precession is

$$\boldsymbol{\Omega} = -\gamma \mathbf{B}.$$

Larmor frequency

For a orbital angular momentum $\mathbf{L}$, $\gamma = \frac{q}{2m}$

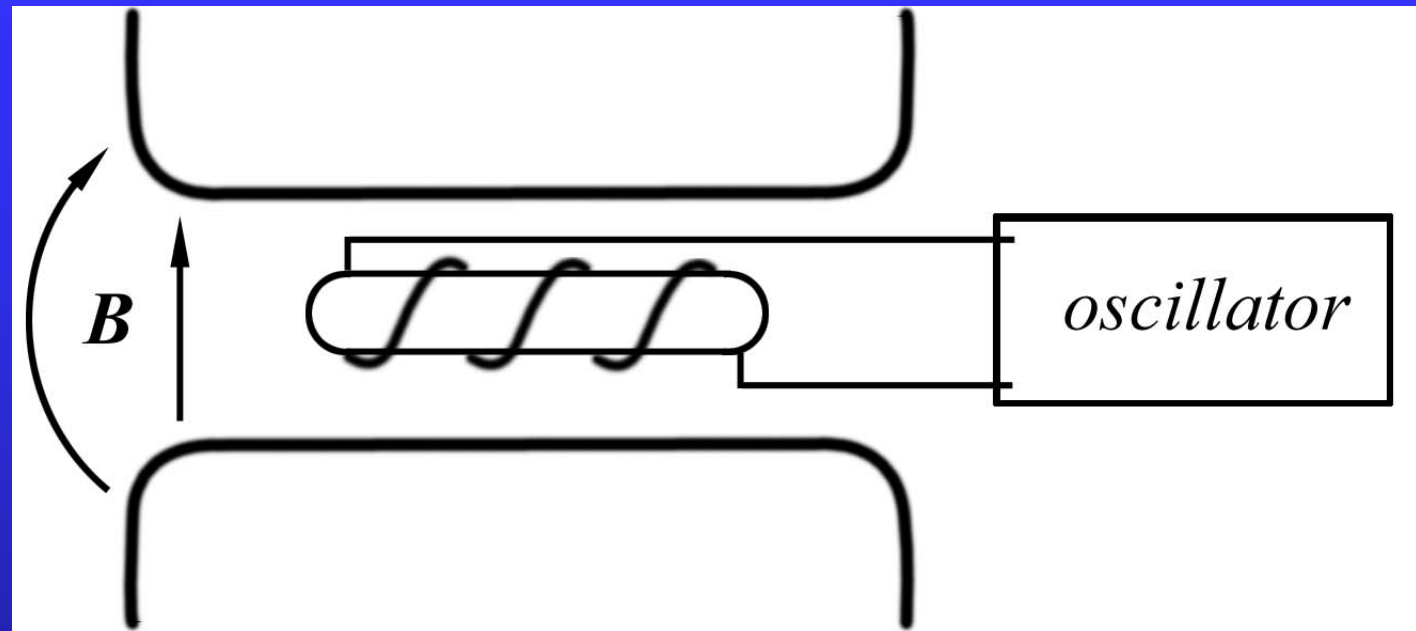
$$\boldsymbol{\Omega} = -\frac{q}{2m} \mathbf{B}.$$

one-half of the
cyclotron frequency

For example, a proton in a 0.50T magnetic field, the frequency of precession is

$$f_p = \frac{\Omega_p}{2\pi} = \frac{\mu_p B}{2\pi \hbar/2} \approx 21\text{MHz}.$$

Nuclear magnetic resonance (NMR)



$$\frac{d\mathbf{S}}{dt} = \boldsymbol{\mu} \times \mathbf{B},$$

$$\frac{d\boldsymbol{\mu}}{dt} = \gamma \boldsymbol{\mu} \times \mathbf{B} = -\gamma \mathbf{B} \times \boldsymbol{\mu}.$$

$$\boldsymbol{\Omega} = -\gamma \mathbf{B}. \quad \text{precession}$$

Assume $\mathbf{B} = B\mathbf{e}_z$, the equation of precessional motion can be written as

$$\left\{ \begin{array}{l} \frac{d\mu_x}{dt} = \gamma B \mu_y, \\ \frac{d\mu_y}{dt} = -\gamma B \mu_x, \\ \frac{d\mu_z}{dt} = 0. \end{array} \right.$$

The solution is

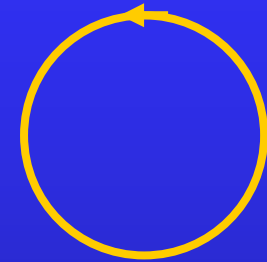
$$\left\{ \begin{array}{l} \mu_x = \mu_{\perp} \cos(|\Omega|t + \delta), \\ \mu_y = -\mu_{\perp} \operatorname{sgn}(\gamma) \sin(|\Omega|t + \delta), \\ \mu_{\parallel} = \mu_{\parallel} \end{array} \right. \quad \begin{array}{l} \gamma > 0, \text{ clockwise} \\ \gamma < 0, \text{ anti-clockwise} \end{array}$$

$$\mu_{\parallel} = \mu_{\parallel} \quad \Omega^2 = \gamma^2 B^2.$$

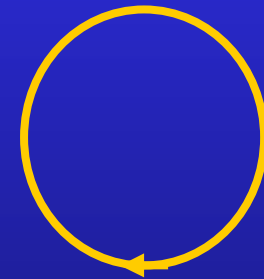
$$\mathbf{B}' = -B' \cos(\omega t + \delta') \mathbf{e}_x$$



$$= \frac{1}{2} B' \cos(\omega t + \delta') \mathbf{e}_x + \frac{1}{2} B' \sin(\omega t + \delta') \mathbf{e}_y$$



$$+ \frac{1}{2} B' \cos(\omega t + \delta') \mathbf{e}_x - \frac{1}{2} B' \sin(\omega t + \delta') \mathbf{e}_y$$



$$\omega = |\Omega| = |\gamma|B \quad \text{resonance}$$

The magnetic moment will flip from up to down.

$$\mu \rightarrow -\mu$$

The energy absorbed is $2\mu \cdot B$.
(from the oscillating field)

By tuning the frequency ω , the signal of spin flip can be detected. Then Ω or γ can be measured.

Applications of NMR

- The measurement of the magnetic moments of nuclei.

$$\begin{array}{ccc} \text{neutron} & \gamma_n & \longrightarrow \mu_n \\ \text{proton} & \gamma_p & \mu_p \end{array}$$

- In the study of materials (including animal tissues), the γ s are known, but the magnetic field contains a contribution from the electrons and nuclei of the material. Therefore, the measurement of B gives information about crystal lattices and about organic material.

- In medical applications, NMR is known as the magnetic resonance imaging(MRI). In the case of medical diagnostics, MRI works primarily on hydrogen atoms.

fat

muscle

tumors

nerve tissue

bones

have different concentrations of H.

effective for neural disease

e. g. multiple sclerosis

(多发性脑脊髓硬化症)

a disease of pituitary

F. Bloch and E. M. Purcell won 1952 Nobel Prize for the development of new methods for nuclear magnetic precision measurements and the discoveries in connection therewith.

Atom and ion have magnetic moments

$$\mu = \sum_i (\mu_L + \mu_S)$$

Some have permanent moment

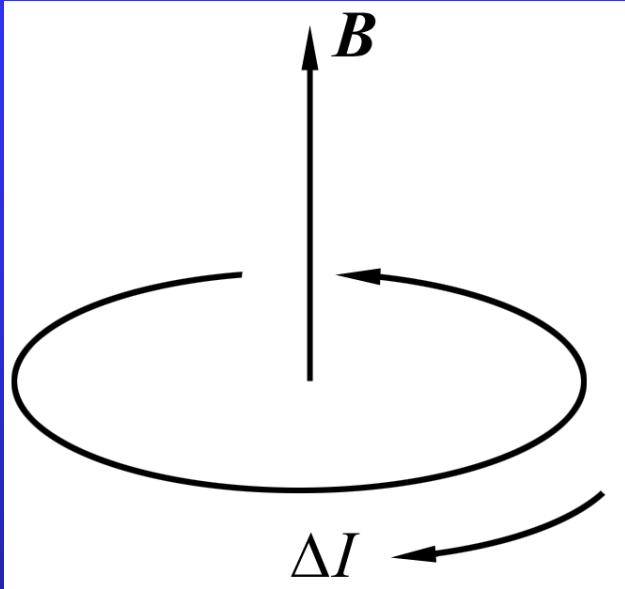
- transition element Mn^{++} ,
- rare earth (51_71) Gd^{+++} ,
- actinide element (89_103) U^{++++} ,
- Al,
- $\text{CrK}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$ (chromium potassium alum).

unfilled (especial inner) shell(s)
the orientation of magnetic moment obeys a
statistical distribution

Others have **induced moment**

- the inert gases He, Ne, Ar, Kr, Xe,
- the alkali Li^+ , Na^+ , K^+ , Rb^+ , Cs^+
- halide F^- , Cl^- , Br^- , I^- , At^-
- Cu, Ag, Au, Hg, CO_2 , H_2 , N_2 ,
- diamond.

Diamagnetism



In accordance with Lenz's law, the induced magnetic moment is anti parallel to the external field.

$$2\pi\rho E_{\phi} = -\pi\rho^2 \frac{dB}{dt}$$

$$M = |\rho \times (-e)E_{\phi}\mathbf{e}_{\phi}| = \frac{1}{2}e\rho^2 \frac{dB}{dt}$$

$$\frac{dL}{dt} = \frac{1}{2}e\rho^2 \frac{dB}{dt}$$

$$dL = \frac{1}{2}e\rho^2 dB$$

$$\Delta L = \frac{1}{2}e\rho^2 (B - 0)$$

Induced moment

$$\Delta\mu = -\frac{e}{2m}\Delta L = -\frac{1}{4m}e^2\rho^2 B$$

replace ρ^2 by $x^2 + y^2$, and set

$$\overline{x^2} = \overline{y^2} = \overline{z^2} = \frac{1}{3}\overline{r^2}$$

$$\Delta\mu = -\frac{e^2}{6m}\overline{r^2} B$$

This result happens to be correct even in quantum mechanics. Usually induced moment is much smaller than permanent moment.

Superconductor

Meissner effect

complete diamagnetism

Assignment: 19.1(a)(b),

19.2 Magnetization

The **magnetization(vector)** of a material is defined as

$$\mathbf{M} = \lim_{\Delta V \rightarrow 0} \frac{\sum \mu}{\Delta V}.$$

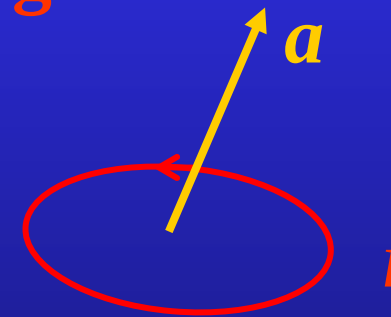
$$\mathbf{M} \rightarrow I_M$$

the **magnetizing current**

Model: $\mu = a i$

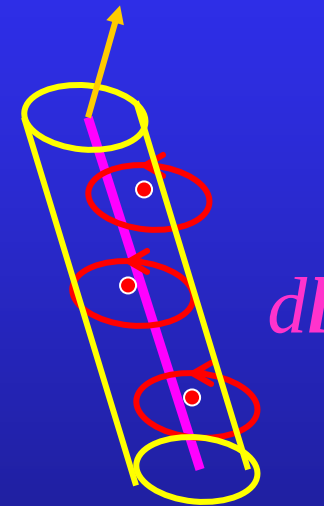
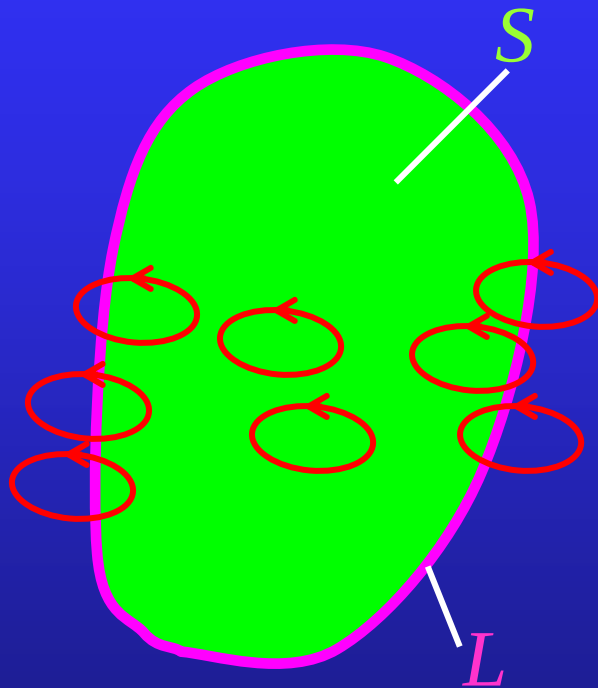
$$\mathbf{M} = n i \mathbf{a}.$$

number of atoms(molecules)
per unit volume



the molecular
current ring

Only the molecular current rings linked by the L contribute the magnetizing current through the surface S .



The number of molecular current rings linked by L is

$$\oint_L n \mathbf{a} \cdot d\mathbf{l}.$$

Thus, the magnetizing current through S is

$$I_M = \oint_L i n \mathbf{a} \cdot d\mathbf{l} = \oint_L \mathbf{M} \cdot d\mathbf{l}.$$

Using Stokes' theorem,

$$I_M = \underbrace{\int_S \mathbf{j}_M \cdot d\mathbf{S}} = \oint_L \mathbf{M} \cdot d\mathbf{l} = \underbrace{\int_S \nabla \times \mathbf{M} \cdot d\mathbf{S}},$$

we have

$$\mathbf{j}_M = \nabla \times \mathbf{M}.$$

Magnetic field H

Taking the magnetizing current into account, the Ampere's law becomes

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 (I_{\text{free}} + I_M)$$
$$= \mu_0 (I_{\text{free}} + \oint \mathbf{M} \cdot d\mathbf{l}).$$

The above equation can be rewritten as

$$\oint \left(\frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \right) \cdot d\mathbf{l} = I_{\text{free}}.$$

Introducing the magnetic field $\mathbf{H}$, defined by

$$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M},$$

the Ampere's law can then be rewritten as

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{free}}.$$

the unit of $\mathbf{H}$: $\text{A} \cdot \text{m}^{-1}$

CGS: oersted (Oe)

$$1 \text{ Oe} = \frac{10^3}{4\pi} \text{ A} \cdot \text{m}^{-1}$$

$\mathbf{B}$ the magnetic induction (intensity)

$\mathbf{H}$ the magnetic field (intensity)

$\mathbf{M}$ the magnetization (vector)

- magnetic field H is related to true current.
- magnetic induction B is related to all “currents”

For a paramagnetic or diamagnetic material and relatively weak field, we have

$$\mathbf{M} = \chi_m \mathbf{H}$$

where the quantity χ_m is called the magnetic susceptibility.

$\chi_m > 0$ for paramagnetic materials

$\chi_m < 0$ for diamagnetic materials

Therefore, the relations among $\mathbf{B}$, $\mathbf{H}$ and $\mathbf{M}$ are

$$\begin{aligned}\mathbf{B} &= \mu_0 \mathbf{H} + \mu_0 \mathbf{M} \\ &= \mu_0 (1 + \chi_m) \mathbf{H} \\ &= \mu_0 \mu_r \mathbf{H}. \\ &= \mu \mathbf{H}.\end{aligned}$$

where μ is called the magnetic permeability, and μ_r is the relative magnetic permeability.

The magnetic energy

$$\oint \mathbf{H} \cdot d\mathbf{l} = Hx = nxI.$$

$$H = nI.$$

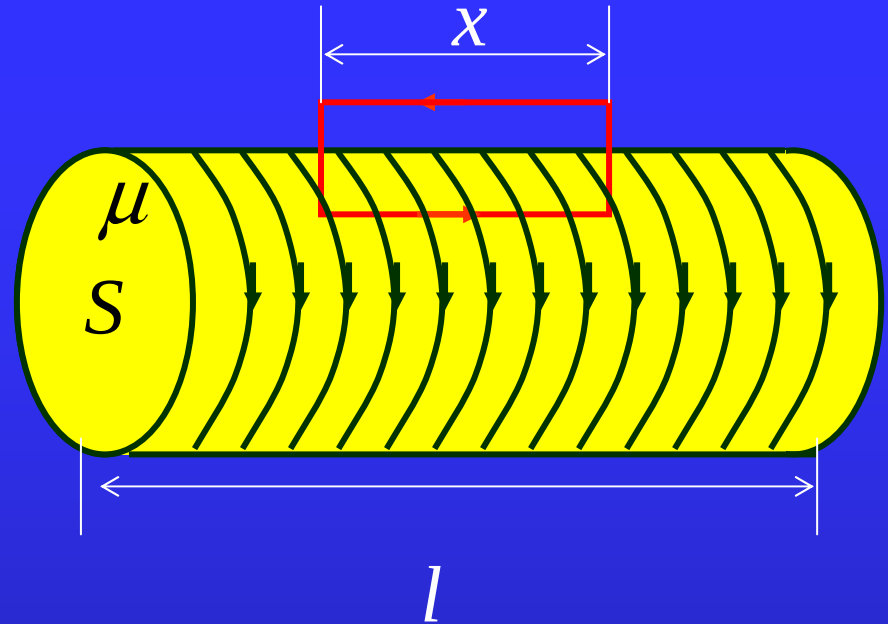
$$B = \mu H = n\mu I.$$

$$\Phi_m = nlBS = n^2\mu lSI. \Rightarrow L = n^2\mu lS.$$

$$U_m = \frac{1}{2}LI^2 = \frac{1}{2}n^2\mu lS \left(\frac{B}{n\mu} \right)^2 = \frac{B^2}{2\mu}lS.$$

$$u_m = \frac{B^2}{2\mu}.$$

$$u_m = \frac{1}{2} \mathbf{B} \cdot \mathbf{H}.$$



Using the relation

$$\mathbf{B} = \mu_0 \mathbf{H} + \mu_0 \mathbf{M},$$

the expression of magnetic energy density can be rewritten as

$$u_m = \frac{1}{2} \mathbf{B} \cdot \mathbf{H} = \frac{1}{2} \mu_0 H^2 + \frac{1}{2} \mu_0 \mathbf{H} \cdot \mathbf{M}.$$

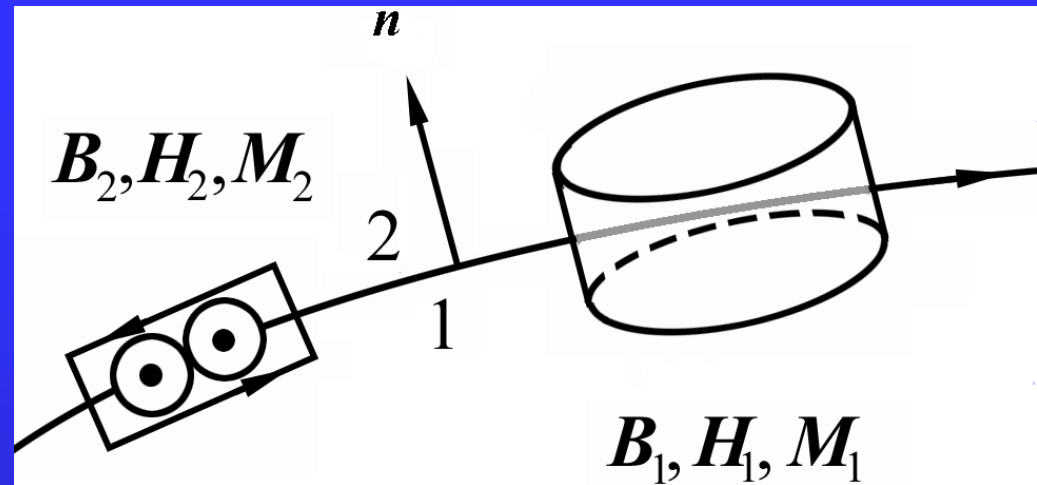
the energy density of
the magnetic field
in the vacuum

potential energy of
the material in the
magnetic field

Boundary conditions

$$\oint \mathbf{B} \cdot d\mathbf{S} = 0$$

$$(\mathbf{B}_2 - \mathbf{B}_1) \cdot \mathbf{n} S = 0$$



the boundary condition for the normal component

$$B_{2n} = B_{1n}.$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{free}}$$

the surface current
on the boundary

If the longer side of the rectangular loop is orthogonal to the surface current density α_{free} , we have

$$(H_{2t} - H_{1t})\Delta l = \alpha_{\text{free}}\Delta l \Rightarrow H_{2t} - H_{1t} = \alpha_{\text{free}}.$$

boundary condition for the tangential component

If the surface current density is zero, the tangential boundary condition becomes

$$H_{2t} = H_{1t}.$$

19.3 Magnetism of materials

paramagnetic

diamagnetic

ferromagnetic

antiferromagnetic

ferrimagnetic

The susceptibility of the diamagnetic and paramagnetic substances

diamagnetic materials

$$\Delta\mu = -\sum_i \frac{e^2}{6m} \overline{r_i^2} \mathbf{B}, \quad \mathbf{M} = n\Delta\mu \equiv \chi_m \mathbf{H}.$$

$$\chi_m = -\frac{n\mu_0 e^2}{6m} \sum_i \overline{r_i^2} < 0, \quad \chi_m \sim -10^{-5}.$$

Sodium	-2.4×10^{-6}
Copper	-1.0×10^{-5}
Bismuth	-1.7×10^{-5}
Diamond	-3.2×10^{-5}

For most substance,
diamagnetism is masked
by the paramagnetism.

paramagnetic materials

$$\mathbf{M} = \frac{N\langle\boldsymbol{\mu}\rangle}{V} = n\langle\boldsymbol{\mu}\rangle.$$

$$\langle\boldsymbol{\mu}\rangle = \frac{\int \boldsymbol{\mu} \exp[-U/(k_{\text{B}}T)] d\Omega}{\int \exp[-U/(k_{\text{B}}T)] d\Omega}$$

$$= k\mu_1 \frac{\int_{-1}^1 \cos\theta \exp[\mu_1 B \cos\theta/(k_{\text{B}}T)] d\cos\theta}{\int_{-1}^1 \exp[\mu_1 B \cos\theta/(k_{\text{B}}T)] d\cos\theta}$$

$$\approx \frac{1}{3} \mu_1 \eta k \quad \left(\eta \equiv \frac{\mu_1 B}{k_{\text{B}}T} \right)$$

$$\mu_1 B \ll k_{\text{B}}T$$

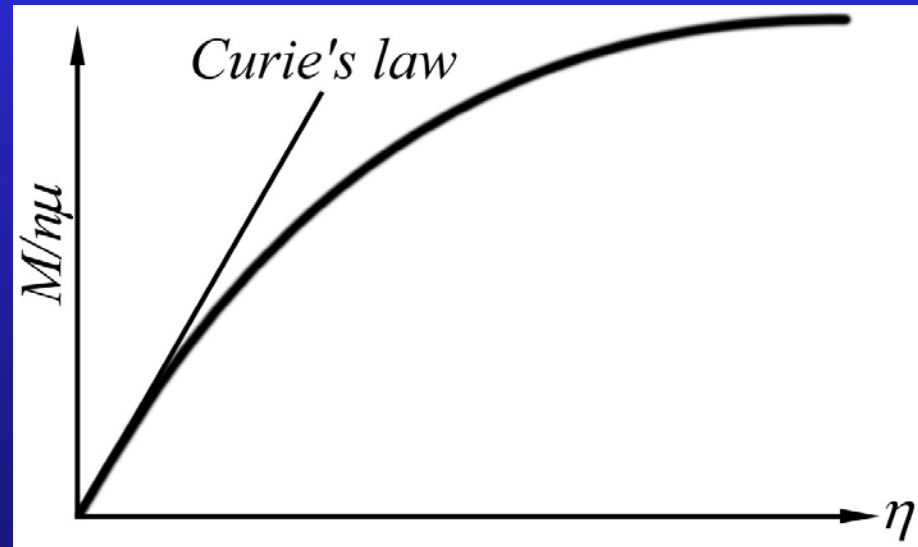
$$\mathbf{M} = n\langle\boldsymbol{\mu}\rangle \approx \frac{1}{3} \frac{n\mu_1^2 \mathbf{B}}{k_{\text{B}}T} \approx \frac{1}{3} \frac{n\mu_1^2 \mu_0 \mathbf{H}}{k_{\text{B}}T}.$$

$$\chi_m = \frac{1}{3} \frac{n\mu_1^2 \mu_0}{k_{\text{B}} T}.$$

Curie's law

$$M = C \frac{B_{\text{eff}}}{T}$$

$$\chi_m \sim \frac{1}{T}$$



Example 19.1 Susceptibility of free-electron gas

$$\chi_m = \frac{1}{3} \frac{n \mu_1^2 \mu_0}{k_B T} - \frac{ne^2 \mu_0}{6m} \overline{r^2} = 0$$

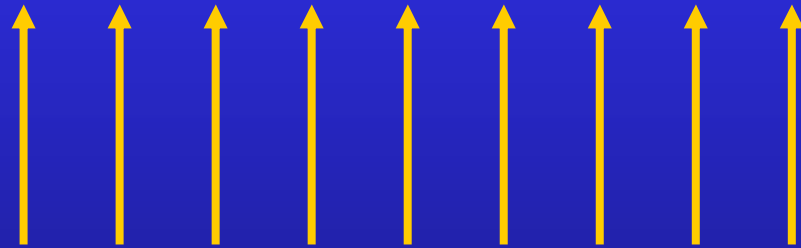
$$\begin{aligned} \mu_1^2 &= \left(-\frac{e}{2m} L \right)^2 \\ &= \frac{e^2}{2m} \frac{L^2}{2mr^2} \overline{r^2} = \frac{e^2}{2m} 2 \cdot \frac{1}{2} k_B T \cdot \overline{r^2}. \end{aligned}$$

The ferromagnetic and antiferromagnetic substances

The exchange interaction for two nearest neighbors is

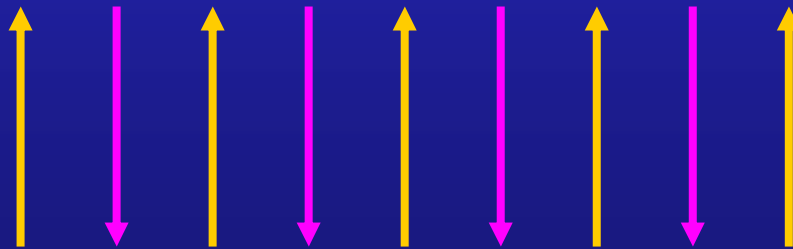
$$U_{\text{ex}} = -JS_i \cdot S_j.$$

$J > 0$



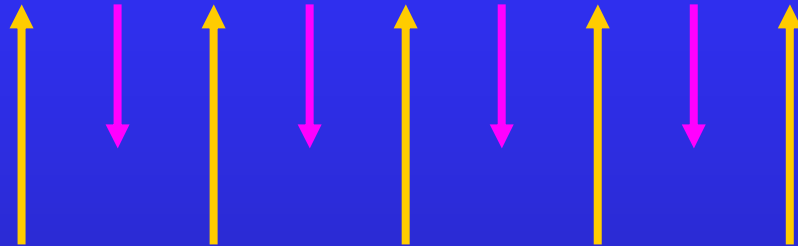
ferromagnetic

$J < 0$



antiferromagnetic

$$U_{\text{ex}} = -JS_i^a \cdot S_j^b. \quad J < 0, \quad S^a \neq S^b.$$



ferrimagnetic

Spontaneous magnetization

Curie temperature T_C

below T_C

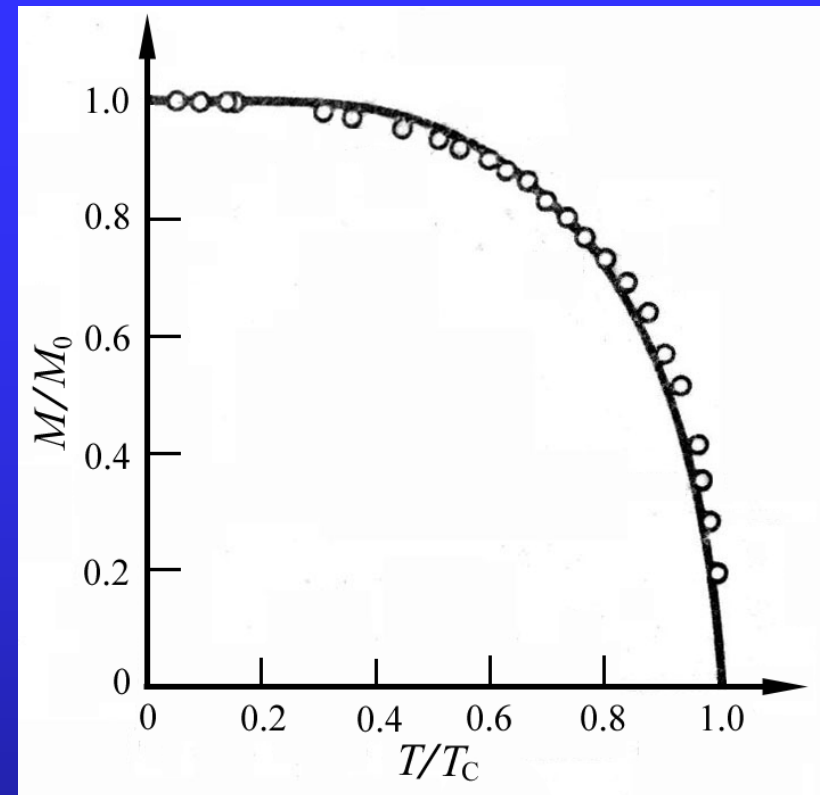
Spontaneous
magnetization

above T_C

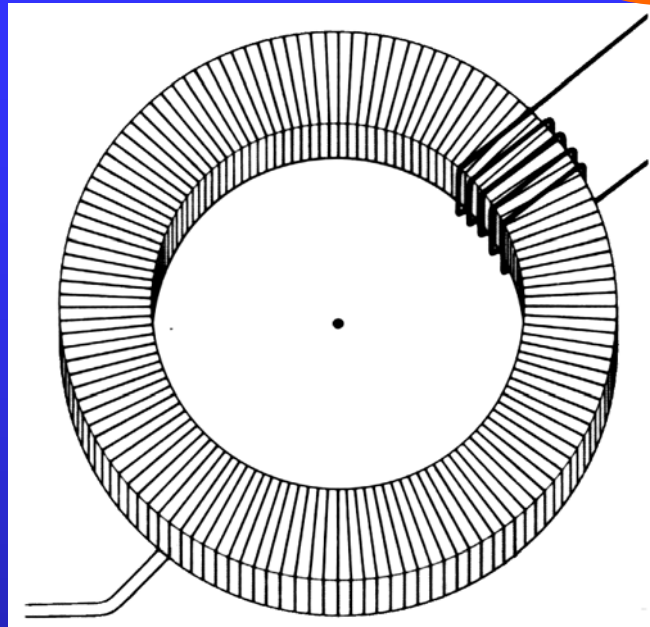
$$M = \frac{C}{T - T_C} H$$

Curie temperature T_C —ferromagnetic-paramagnetic
transition temperature

Fe(iron)	Co(cobalt)	Ni(nickel)	Gd(gadolinium)	Dy(dysprosium)
1043K	1393K	631K	289K	105K



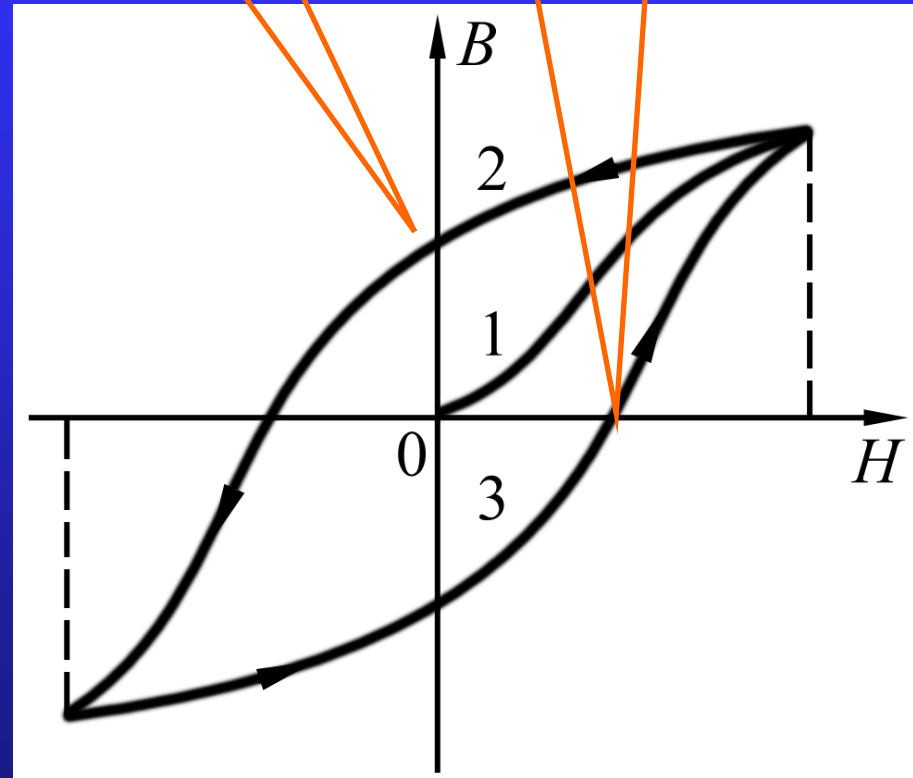
Hysteresis loop



Roland ring

remanence

coercive force H_C



H —external magnetic field, B —internal magnetic field

large H_C ($10^4 \sim 10^6 \text{ Am}^{-1}$) magnetically hard

It's difficult to magnetize, but once magnetized, it's difficult to demagnetize.

These materials can be used as permanent magnets or the magnetic recording material.

small H_C ($\sim 1 \text{ Am}^{-1}$) magnetically soft
transformer core

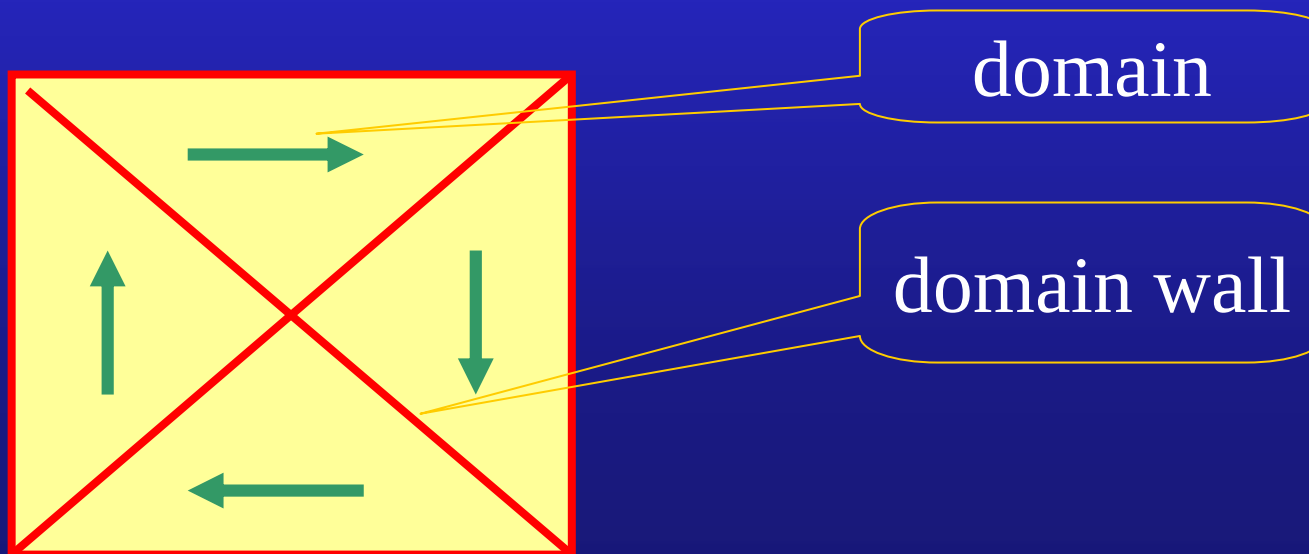
pure iron—soft

add carbon or tungsten(wolfram) —→ hard

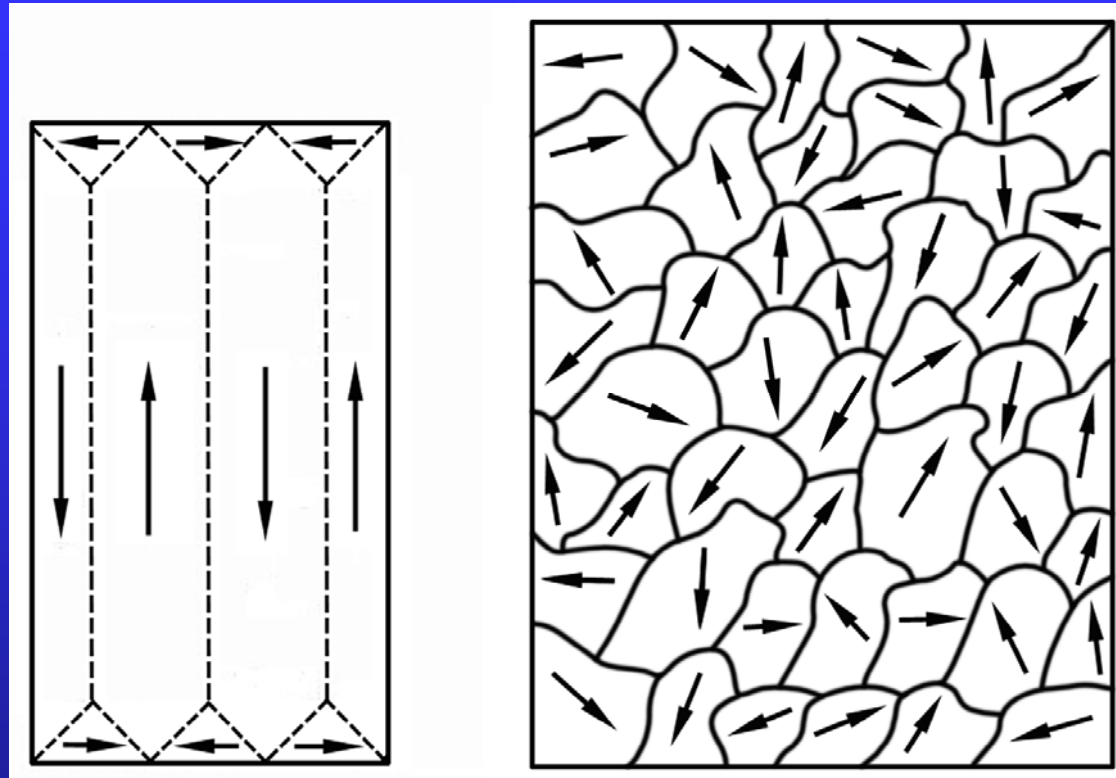
Magnetic domains

The ferromagnetic substances don't show bulky magnetization until they are magnetized. The reason is the existence of magnetic domains. In different domains, the orientations of the magnetization are different.

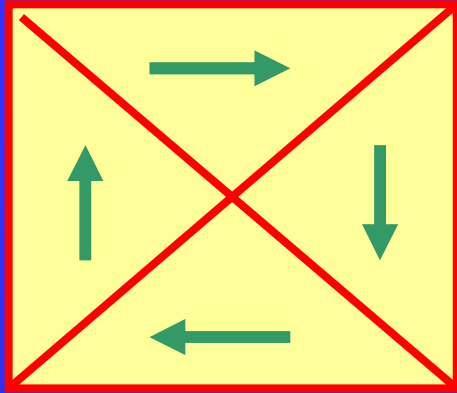
colloidal suspension+ Fe_3O_4 powder



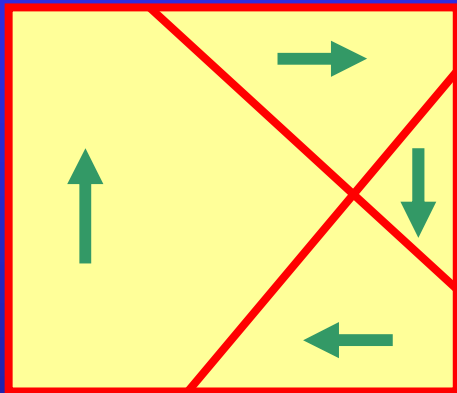
Magnetic domains of single crystal and polycrystal



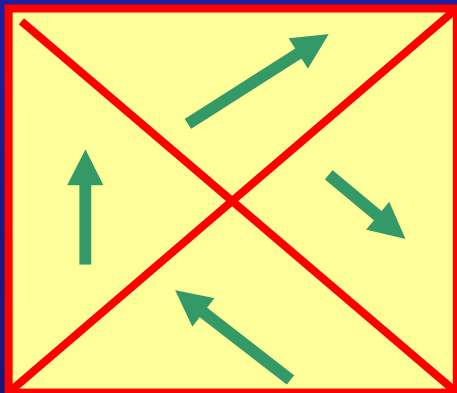
$10^{17} \sim 10^{21}$ atoms, size: $10^{-12} \sim 10^{-8} \text{m}^3$,
scale: 0.1~1mm.



unmagnetized



magnetization by domain
growth



magnetization by
reorientation

Assignment: 19.4 or 19.5, 19.8